

Predicting the severity of traffic accidents

Big Data — Final Project
Innopolis University

Team 31

Anna Belyakova	<code>a.belyakova@innopolis.university</code>
Sofia Pushkareva	<code>s.pushkareva@innopolis.university</code>
Daria Alexandrova	<code>d.alexandrova@innopolis.university</code>
Aisylu Fattakhova	<code>a.fattakhova@innopolis.university</code>

Introduction

Business objectives

Data Description

Data Characteristics

Architecture of data pipeline

The input and the output for each stage

Data preparation

ER diagram

Some samples from the database

Creating Hive tables and preparing the data for analysis

Data analysis

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Training and fine-tuning

Evaluation

Data presentation

The description of the dashboard

Description of each chart

Your findings

Conclusion

Summary of the report

Reflections on own work

Challenges and difficulties

Recommendations

The table of contributions of each team member

Project tasks	Task description	Anna Belyakova	Sofia Pushkareva	Daria Alexandrova	Aisylu Fattakhova	Deliverables	Average hours spent
Data extraction	Collect the US Accidents dataset and stage a representative sample for the pipeline.	25%	25%	25%	25%	US_Accidents_March23.csv	—
Data cleaning & profiling	Chunked preprocessing; deduplication; coordinate and boolean cleanup; quality report.	25%	25%	25%	25%	*_clean.csv; output/data_quality_stage1.txt	—
PostgreSQL & Sqoop	DDL, bulk load, export to HDFS/Avro for Hive ingestion.	25%	25%	25%	25%	sql/*.sql; HDFS warehouse paths	—
Hive & EDA	External/partitioned tables; HiveQL analytics; CSV exports for dashboard.	25%	25%	25%	25%	hive/*.hql; output/q*.csv	—
Spark ML	Feature pipeline; train/tune RF, LinearSVC, Naive Bayes; cross-validation.	25%	25%	25%	25%	models/; evaluation outputs	—
Dashboard & presentation	Superset/Streamlit dashboard; defence slides and storytelling.	25%	25%	25%	25%	Dashboard URL; demo assets	—
Reporting	Write-up of pipeline, results, and reflections (report/main.tex).	25%	25%	25%	25%	This report (PDF)	—